

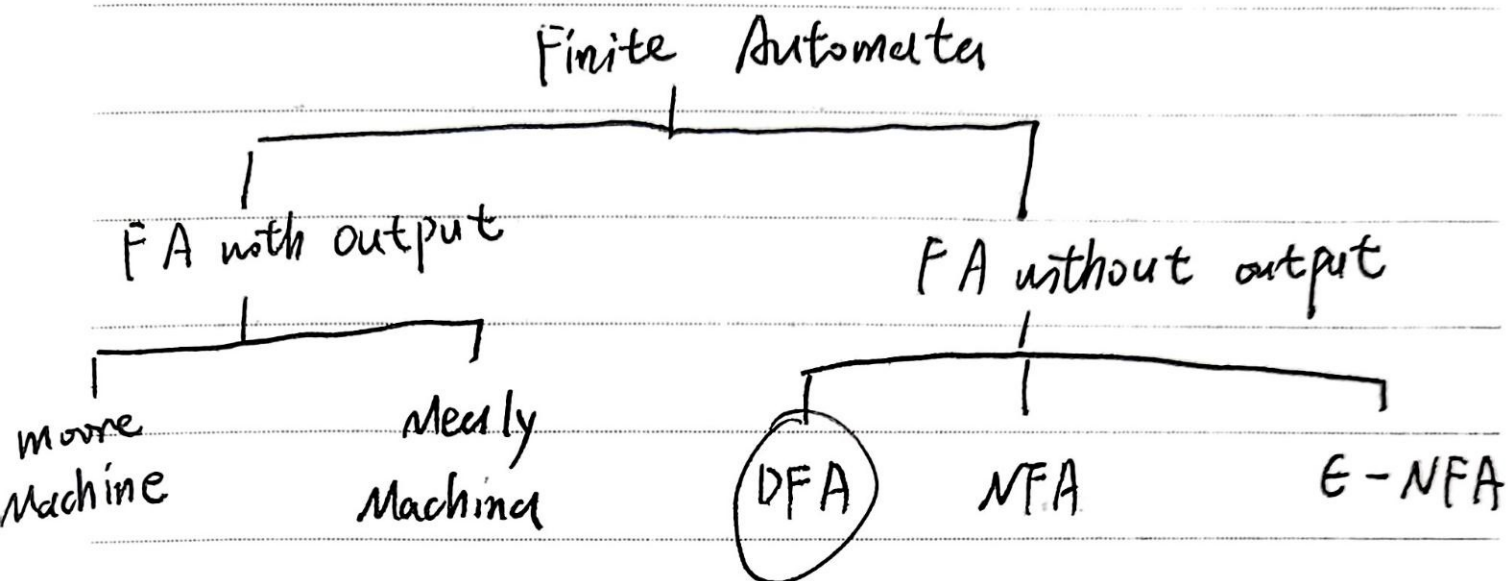


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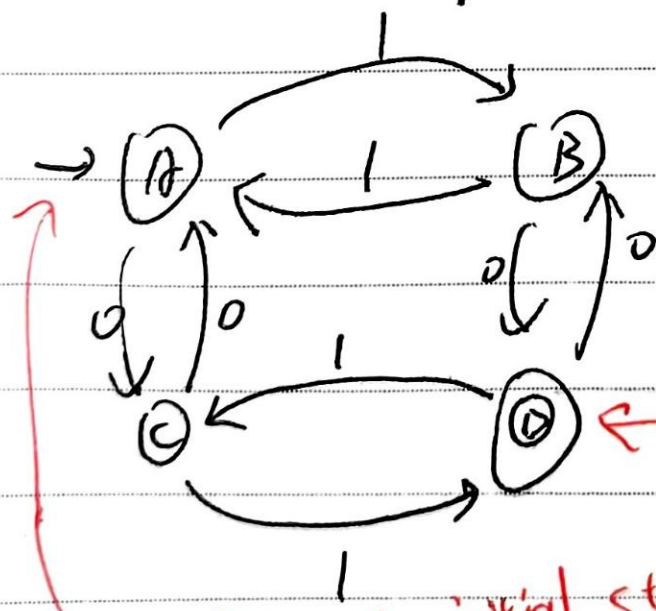
3. Finite State Machine



DFA - Deterministic Finite Automata

- It is the simplest model of computation

- It has a very limited memory



states: A, B, C, D

D is the final state of DFA

means A is the initial state (start state) of DFA

⇒ definition of this DFA $(Q, \Sigma, q_0, F, \delta)$

Q = set of all states (2)
 Σ = inputs
 q_0 = start state / initial state
 F = set of final states,
 δ = transition function from $Q \times \Sigma \rightarrow Q$
 ↑ (del)

for upon ↑ DFA:

$Q = \{A, B, C, D\}$,

$\Sigma = \{0, 1\}$

$q_0 = A$, $F = \{D\}$

δ : (del)

	0	1
A	C	B
B	D	A
C	A	D
D	B	C

$Q, \Sigma, q_0, F, \delta$